

Erin Sun

erinsun328@gmail.com | (949) 346-6769 | Irvine, CA

Education

University of California Santa Barbara

September 2022 - Present

- Bachelor of Science - Mechanical Engineering
 - Minor - Media Arts and Design (expected completion September 2026)
-

Work History

Kingston Technology | Intern

July 2024 - September 2024

- Designed and prototyped packaging alignment solutions in SolidWorks that improved production line efficiency by 70%
 - Fabricated complex parts using 3D printing to manufacture prototypes and run tests
 - Assisted with equipment installation and setup across the company using a variety of hand tools
 - Performed PCB solder removal and rework using solder wick and operated solder application/removal equipment
 - Rotated through various stations along the SMT production line to observe and support manufacturing operations
-

Projects

Xbox controller charging dock

2026 - Present

- Designing a charging dock and external battery pack for Xbox controllers to improve charging accessibility and reduce battery replacement frequency

Senior Capstone | CAD Lead

September 2025 - June 2026

- Developed an adaptive gym tool to assist individuals with cerebral palsy in performing workouts more independently

Soft Robot

January 2026 - March 2026

- CAD-modeled and redesigned a pneumatically actuated soft crawling robot inspired by a published research study using silicone-based fabrication
- Tested pneumatic chambers to evaluate crawling motion and actuator performance

Junior Design | Team Lead, CAD Lead

April 2025 - June 2025

- Developed a smart walking cane incorporating distance sensors and haptic feedback to improve mobility for individuals with visual impairments
- Researched, designed, and produced the handle and housing components using CAD

Course navigating robot

January 2025 - March 2025

- Designed and built a robot using only cardboard to navigate a custom course and successfully deliver a ping pong ball to a target destination
-

Skills

Technical

- CAD & Simulation: SolidWorks (CSWP Certified), Fusion 360, FEA, CAM
- Programming: MATLAB, Python, Arduino IDE
- Manufacturing: CNC mill, lathe, laser cutting, soldering, 3D printing, SMT processes
- Other: Excel, FlashCut, Procreate

Languages

- English (fluent)
- Mandarin Chinese (conversational proficiency)